

Vol 9 Chapter 2 — Crystal Lattices

Keeper (author pass — deep math/physics revision)

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Chapter 2 — Crystal Lattices

Level 1 — one sentence

Crystalline solids have periodic atomic arrangements classified by the 14 Bravais lattices and 230 space groups, giving Bloch's theorem $\psi(\vec{r}) = e^{i\vec{k}\cdot\vec{r}}u(\vec{r})$ as the fundamental solution-form for electrons in periodic potentials, with reciprocal lattices and Brillouin zones structuring all of solid-state band theory.

Level 2 — graduate-physicist precision

2.1 The 14 Bravais lattices

A Bravais lattice is an infinite array of points generated by primitive translation vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3$:

$$\vec{R} = n_1\vec{a}_1 + n_2\vec{a}_2 + n_3\vec{a}_3, \quad n_i \in \mathbb{Z}$$

In 3D, 14 distinct Bravais lattices (Bravais 1849): - 7 crystal systems (triclinic, monoclinic, orthorhombic, tetragonal, cubic, hexagonal, trigonal) - Plus various centerings (P primitive, I body-centered, F face-centered, C base-centered)

Specific examples: - Simple cubic (SC): primitive vectors along $\hat{x}, \hat{y}, \hat{z}$ - Body-centered cubic (BCC): atoms at corners + cube center (Fe, Na) - Face-centered cubic (FCC): atoms at corners + face centers (Cu, Al, Au)

K50 in BST audit chain investigates Bravais lattices as candidate L1 ESTABLISHED source.

2.2 The 230 space groups

Adding rotation, reflection, glide, screw symmetries to the 14 Bravais lattices gives 230 space groups in 3D — the complete classification of crystallographic symmetry (Schönflies 1891, Fedorov 1891). All real crystals belong to one of these.

2.3 Reciprocal lattice and Brillouin zones

The reciprocal lattice has primitive vectors $\vec{b}_i$ satisfying $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$. Explicit: $\vec{b}_1 = 2\pi(\vec{a}_2 \times \vec{a}_3)/[\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)]$, cyclic.

Reciprocal lattice points $\vec{G} = m_1\vec{b}_1 + m_2\vec{b}_2 + m_3\vec{b}_3$ satisfy $e^{i\vec{G}\cdot\vec{R}} = 1$ for all direct-lattice vectors $\vec{R}$.

Brillouin zone = Wigner-Seitz cell of reciprocal lattice — the natural domain for crystal momentum $\vec{k}$.

2.4 Bloch's theorem

For a single electron in a periodic potential $V(\vec{r} + \vec{R}) = V(\vec{r})$:

$$\psi_{n\vec{k}}(\vec{r}) = e^{i\vec{k}\cdot\vec{r}} u_{n\vec{k}}(\vec{r})$$

with $u_{n\vec{k}}(\vec{r} + \vec{R}) = u_{n\vec{k}}(\vec{r})$ periodic. The wave function has plane-wave envelope $e^{i\vec{k}\cdot\vec{r}}$ modulated by a lattice-periodic function u .

n is the band index; $\vec{k}$ is crystal momentum in first Brillouin zone.

Substrate reading: Bloch's theorem expresses the substrate's K-type structure on a periodic lattice — the K-type is labeled by band n and Brillouin-zone-restricted momentum $\vec{k}$.

2.5 X-ray diffraction and Bragg's law

X-rays diffract from crystals at angles satisfying $2d \sin \theta = n\lambda$ (Bragg's law, 1912). The reciprocal-lattice formulation: scattering vector $\vec{q} = \vec{k}_{\text{out}} - \vec{k}_{\text{in}}$ must equal a reciprocal-lattice vector $\vec{G}$ for constructive interference.

Crystallography techniques: X-ray, neutron, electron diffraction. Determine atomic positions to $<0.01 \text{ \AA}$ precision. Foundation for materials science, structural biology (DNA structure, protein structures all from crystallography).

2.6 BaTiO₃ 137-plane (preview Ch 8)

BaTiO₃ has a perovskite structure (Pm $\bar{3}$ m space group at high T, P4mm at room T after ferroelectric transition). The "137-plane" crystallographic configuration is the specific plane orientation where the substrate's $N_{\text{max}} = 137$ engages. Ch 8 develops the BST eigentone prediction.

2.7 K-audit anchors

- K50 (BST audit chain): Bravais Root candidate
- Vol 0 Ch 5: $N_{\text{max}} = 137$ connection to crystallography
- Ch 8 of this volume: BaTiO₃ 137-plane

Level 3 — 5th-grader accessibility

Crystalline solids have periodic atomic arrangements — 14 Bravais lattices in 3D (simple cubic, body-centered cubic, face-centered cubic, ...), 230 space groups total. Bloch's theorem: electrons in a periodic potential have wave functions $\psi = e^{i\vec{k}\cdot\vec{r}} u(\vec{r})$ — plane wave modulated by a lattice-periodic part. Brillouin zone: the natural domain for crystal momentum. X-ray diffraction: Bragg's law $2d \sin \theta = n\lambda$. Crystals reveal their atomic structure via diffraction patterns. Foundation for everything in solid-state physics. BST connection: the 137-plane of BaTiO₃ engages substrate's $N_{\text{max}} = 137$ for a BST eigentone prediction (Ch 8, ~\$25K test).

What comes next

Chapter 3 develops electron band structure.

Where to look this up

- Ashcroft and Mermin Ch 4-7
- Kittel, *Introduction to Solid State Physics*
- BST: K50 audit; Vol 0 Ch 5